	QUESTION	A
1	What is the relationship between sample size and the standard error of the mean?	*The standard error decreases as the sample size increases.
2	Which of the following statements is true?	*The standard error is calculated solely from sample attributes.
3	What is one of the distinctions between a population parameter and a sample statistic?	*A sample statistic changes each time you try to measure it, but a population parameter remains fixed.
4	A magazine printed a survey in its monthly issue and asked readers to fill it out and send it in. Over 1000 readers did so. Choose the appropriate name this type of sampling.	*a self-selected sample.
5	Which of the following would be most likely to produce selection bias in a survey?	*Using a random sample of students at a university to estimate the proportion of people who think the legal drinking age should be lowered.
6	Which one of these statistics is unaffected by outliers?	*Interquartile range
7	Which of the following would indicate that a dataset is not bell-shaped?	*The mean is much smaller than the median.
8	What is the effect of an outlier on the value of a correlation coefficient?	*An outlier might either decrease or increase a correlation coefficient, depending on where it is in relation to the other points.
9	Pick the choice that best completes the following sentence. If a relationship between two variables is called statistically significant, it means the investigators think the variables are….	*related in the population represented by the sample.
10	A random sample of 5 mosquitos is sampled. The number of mosquitos carrying the West Nile Virus in the sample is an example of which random variable?	*binomial

11	A data manager needs to learn if hospital staff are getting fewer hours of break than the 12 hours stated on their working contract. After she collects data from a sample of some staff, she is going to conduct a test of a hypothesis. She should use…	*a one tailed test with an alternative to the left.
12	A researcher is studying patients in Rwandan Hospitals. She takes a sample of 400 patients from 10 hospitals. The average age of all hospital patients in Rwanda is….	*a parameter
13	Half of the observations in a data set are greater than the	*median
14	In a random sample of 28 families, the average weekly food expense was $95.60 with a standarddeviation of $22.50. Determine whether a normal distribution or a t-whether neither of these can be used to construct a confidence interval. Assume	*Use the t-distribution.
15	Edit program is required to	*detect errors in input data
16	Data rejected by edit program are	*corrected and re- entered
17	The choice of data analysis tools depend on the research design and	*research methodology.
18	The key steps of an evidence-based design process includes all of the following EXCEPT	*developing original data collection instruments
19	During the data collection phase, researchers do all of the following EXCEPT	*analyze the data.
20	A scatter plot uses dots to represent values for two different numeric variables. The position of each dot on the horizontal and vertical axis indicates values for an individual data point. Scatter plots are used to observe relationships between variables. Now choose the correct procedures on how you can set up a scatter graph for the table.	*1. Select all the cells in the table with the cursor. 2.Click the Insert tab and select Scatter > Scatter with only Markers to add the graph to the spreadsheet as below. 3. Then you should right-click the chart and select Change Chart Type > X Y (Scatter) > Scatter with only Markers.
21	When the correlation coefficient, r, is close to one:	*there is a strong linear relationship between the two variables

22	As Data manager, what is the right procedure among the proposed Six steps to good questionnaire design	*1. Identify your research aims and the goal of your questionnaire/ 2. Define your target respondents/ 3. Develop questions 4. Choose your question type 5. Design question sequence and overall layout 6. Run a pilot
23	Here we provide you with the key differences between the z-test vs. t-test you must know. Select the best answer	*1. One of the essential conditions for conducting a t-test is that population standard deviation or the variance is unknown/ 2. Z-test is used when the sample size is large, which is n > 30
24	In a study of possible correlation between the height in cm (X) and weight in kg (Y) of newborns, a sample of 40 babies produces a correlation coefficient of r=+0.813 and a regression line with equation of Y=0.34X+19.5. What is the expected weight of an 80 cm tall newborn?	*The expected weight is 46.7kg
25	A population is divided into three strata so that A=5000, B=2,000 and C=3000. The respective standard deviations are s1=15, s2=18 and s3=5. How should a sample size n=84 be allocated to three strata, if we want optimum allocation using disproportionate sampling design?	*The required sample size for the population is allocated as follows: (i) 52 for population A, (ii)25 for population B and (iii) 7 for population C.
26	What if the standard deviation is 0?	*When the standard deviation is zero, there is no spread; that is, the all the data values are equal to each other.
27	Choose among the following answers on the main four drivers of the population dynamics in any region	*Deaths, Births, Immigrants, Emigrants
28	Q41.A.The 2002 Rwanda population census indicated a total population of 8,128,553 while the Fourth (2012) Rwanda Population and Housing Census (RPHC4) established that the population of Rwanda was 10,515,973 residents, of which 5,064,868 were males. Work on the following: Q41.A. Determine the Rwanda population growth rate .	*0.026
29	Q41.C. Calculate the population density for 2002	*Approximatelly 309 populations/Sq Km

30	Q41.D. Calculate the population density for 2012	*Approximatelly 400 populations/Sq Km/
	A survey was conducted in district Y in 2016 and revealed that there is a total population of 389,000 (196,500 males and 192,500 females). The total number of children born alive was 16,400. The survey has also recorded a total of 5835 deaths (3200 males and 2635 females) during the same year. The survey has also identified the following deaths among the population. Given these information: Children < 1 month=370, Children 1 month to 11 months=1100, Children 1 year to 4 years=1865 and Death of mothers during pregnancy=130. Based on the above data provided, calculate the following measures of fertility and mortality: Q47.A. Crude birth rate(CBR).	*0.04215
31		
32	Q47.B. Calculate the Crude death rate(CDR)	*0.015
33	Q47.C. Calculate the infant mortality rate (IMR)	*0.089